********** QUESTIONS AND SOLUTIONS **********

Information for Candidates

Numerical answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

Notation

The following notation is used in this paper:

- The voltage waveform at node X in an AC circuit is denoted by $x(t)$ while the phasor voltage is denoted by X.

- Component and source values are generally given in Ohms, Farads, Henrys, Volts or Amps with the unit symbol omitted. Where an imaginary number is specified, it represents the complex impedance or phasor value.

- Times are in seconds unless otherwise stated.

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Solution

1. a) Using nodal analysis, calculate the voltages at nodes *X* and *Y* of Figure 1a. [6]

KCL at X gives: $\frac{X-20}{30}+\frac{X-(Y-13)}{20}+\frac{X-Y}{10}=0 \quad \Rightarrow \quad 11X-9Y=1.$

KCL at the supernode {Y, Y-13} gives: $\frac{Y-X}{10}+\frac{Y-20}{15}+\frac{Y-13}{10}+\frac{(Y-13)-X}{20}=0$

$\Rightarrow \quad -9X+19Y=197.$

Solving these two equations gives: $X=14\,V\,;Y=17\,V.$

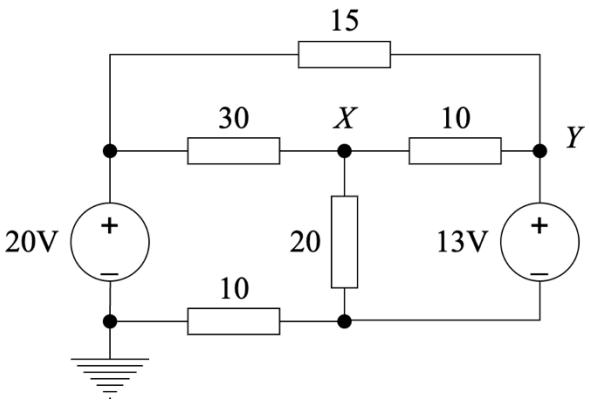


Figure 1a

b) Using the **principle of superposition**, find the Thévenin's equivalent of the circuit between nodes A and B in *Fig. 1b*.

[6]

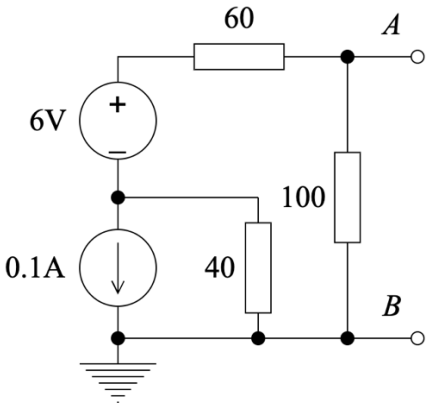
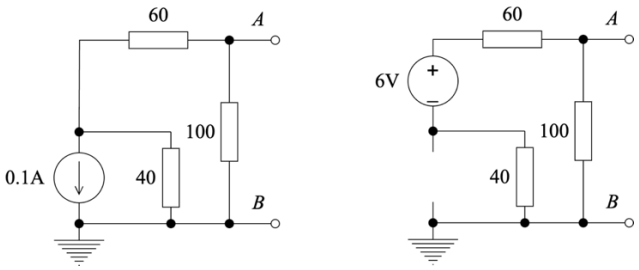


Fig. 1b

Setting the voltage source to zero gives us the first diagram shown in the solution below. Combining 40||(60 + 100) = 32 so X = −0.1 × 32 = −3.2 V. It follows (potential divider) that $V_{AB} = -3.2 \times 100/160 = -2\ V.$

Now setting the current source to zero gives the second diagram and we have a potential divider giving $V_{AB} = 6\times 100/200 = 3V.$

By Superposition, $V_{AB} = V_{Th} = -2+3 = 1V.$



To find R_{Th} we set both sources to zero and find the resultant resistance of 100||(60+40) = 100||100 = 50 Ω.

c) Calculate the complex power associated with the resistor, inductor and source in *Fig. 1c* when $C = 0$. Then, determine the value for the capacitor C that will render the complex power delivered by the source purely real. The source has a frequency of 50 Hz. [8]

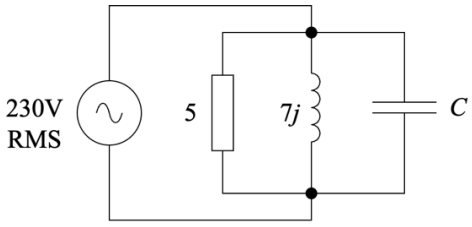


Fig. 1c

If we take the source to have zero phase then its (RMS) phasor is: $V = 230$.
With $C = 0$, *the load impedance is:*
$Z = 5\,||(7j) =$ *3.3108 + j 2.3649 =* *4.06868∠35.5382°*

The current phasor is therefore: $\tilde{\tilde{I}} = \frac{\tilde{V}}{R} + \frac{\tilde{V}}{Z_L} = 46 - j32.857\ A$
$S = \tilde{V}\tilde{I}^* = 10.58 + j\ 7.557\ kVA$
$P = 10.58\ kW$
$Q_L = 7.557\ kVAr$

To render the source power purely real, the admittance of the capacitor needs to cancel out the imaginary part of the RL load admittance, i.e. we require $\omega C = -Im\left\{\frac{1}{Z}\right\}$.

$$Z_C = \frac{1}{j\omega C} = -7j\ \Omega$$

C = 454.73 μF

The parallel capacitor should have apparent power which is opposite to the reactive power of the inductor (S_C = -7.557 j). The new apparent power from the source will be $S = \tilde{V}\tilde{I}^* = 10.58 + j\ 0\ kVA = 10.6\angle 0°\ kVA$

d) For the amplifier of *Fig. 1d*, calculate values of R_B and R_C that set the quiescent value of V_{out} at 5 V with 1.5 mA of collector current. Draw a small signal model and calculate the small signal voltage gain. Do not ignore R_B in your calculations.

[8]

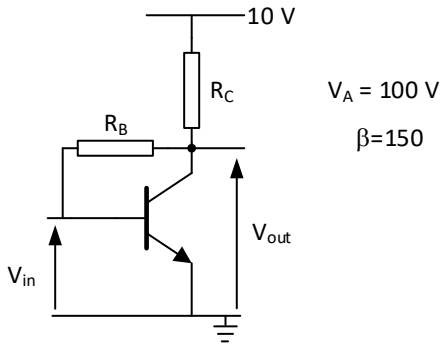
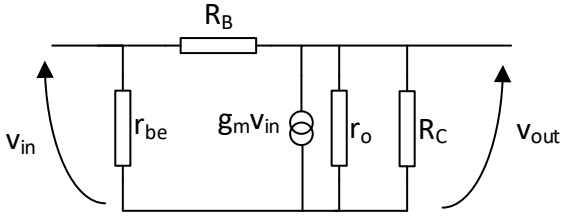


Fig. 1d

In the quiescent state, to obtain 1.5 mA of collector current, a base current of 10 μA is needed. This sets R_B=(5-0.7)/10μ=430kΩ. R_C=(10-5)/1.5m=3.5 kΩ.

Under these conditions, the small signal transistor parameters are:
g_m=1.5m/25m=60 mS
r_o=100/1.5m=66.7 kΩ
r_{be}=150/60m=2.5 kΩ

The small signal model is therefore:



KCL at the output gives:

$$\frac{v_{out}}{R_C} + \frac{v_{out}}{r_o} + \frac{v_{out} - v_{in}}{R_B} + g_m v_{in} = 0$$

Giving:

$$\frac{v_{out}}{v_{in}} = A_v = \frac{\frac{1}{R_B} - g_m}{\frac{1}{R_C} + \frac{1}{r_o} + \frac{1}{R_B}} = \left(\frac{1}{R_B} - g_m\right)(R_B // R_C // r_o)$$

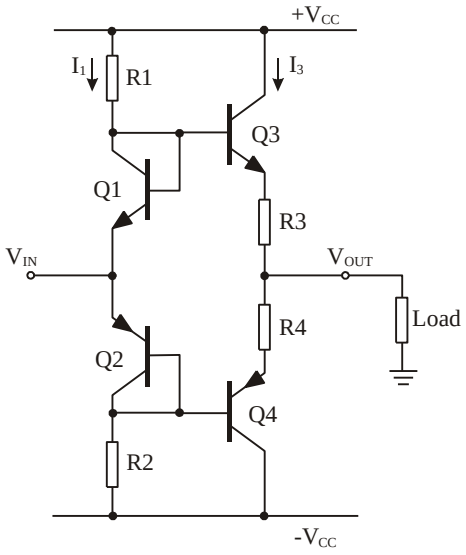
This enumerates to -198 (3 d.p)

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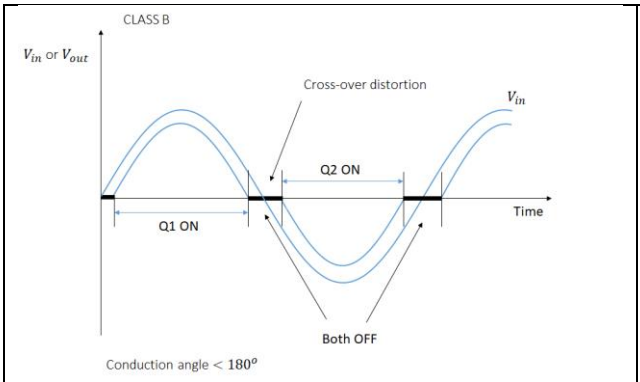
e) Draw a class AB amplifier arrangement, explain its use and state why it is superior to a class B stage. You may wish to draw a sketch graph of the input and output voltage for both class B and class AB stages to explain your answer.

[5]

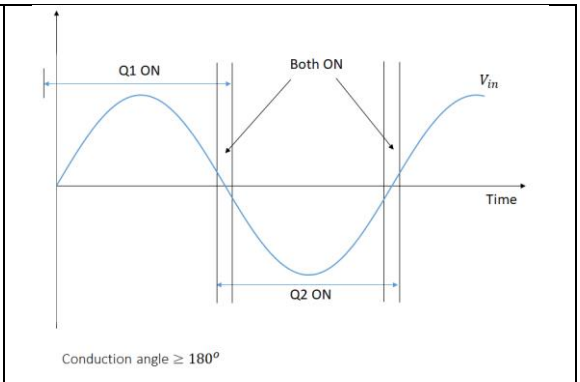
A class AB arrangement is shown below. The voltage gain of the amplifier is 1, but the circuit provides a current (and hence power) gain.



The class B stage exhibits cross over distortion due to the conduction angle of each transistor being less than 180 degrees, but the class AB almost eliminated this distortion by setting the conduction angle to >=180 degrees.



Class B input and output



Class AB input and output

f) A new type of transistor called the Imperial-FET, shown in *Fig. 1f*, has been proposed. It has a circuit symbol shown below. The terminals are called *control* (c), attractor (a) and *repeller* (r). current is considered positive into the attractor and out of the *repeller*. There is no current flow into the control terminal.

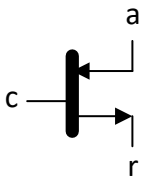


Fig. 1f

The current into the attractor, I_a, is controlled solely by the Control-Repeller Voltage, V_{cr}, by the equation:

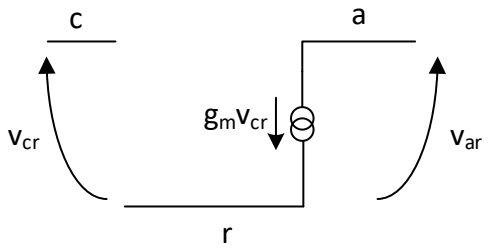
$$I_a = \sin\left(\frac{\pi}{2}V_{cr}\right)$$

Taking inspiration from the small signal MOSFET model, draw a small signal model of the Imperial-FET, and derive any small signal parameters, explaining your method.

[7]

We are told to take inspiration from the MOSFET model (which makes sense because we are told not current flow in the control terminal. We can also see that I_a only depends on V_{cr} and not V_{ar}. Hence the only small signal parameter that is needed is a transconductance.

The transconductance is $\frac{dI_a}{dV_{cr}} = g_{m=}\frac{\pi}{2}cos\left(\frac{\pi}{2}v_{cr}\right)$ and hence the small signal model is:



2. a) Write down the transfer function $\frac{Y}{X}(j\omega)$ for the circuit in *Fig. 2a*. Find the high and low frequency asymptotes of the transfer function to show that the circuit is a high pass filter. [8]

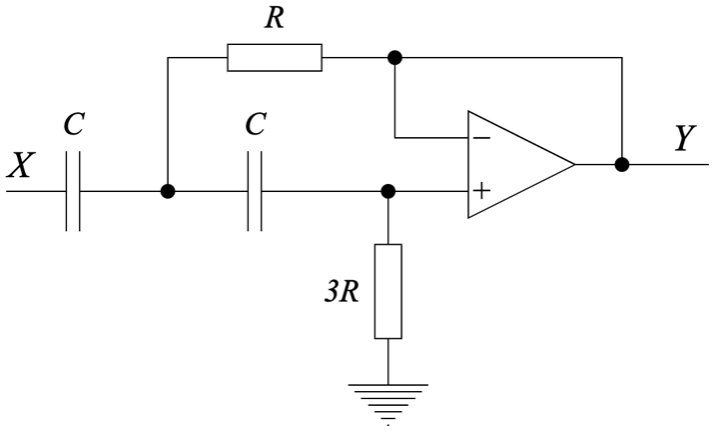


Fig. 2a

Consider the unlabeled node is node Z (as shown in the circuit below)

KCL @Z: $\frac{Z-X}{1/j\omega C} + \frac{Z-Y}{1/j\omega C} + \frac{Z-Y}{R} = 0$

$$\Rightarrow Z(1 + 2j\omega RC) - Y(1 + j\omega RC) = Xj\omega RC$$

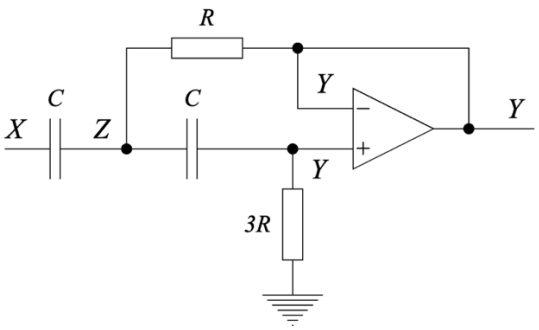
KCL @V_+: $\frac{Y}{3R} + \frac{Y-Z}{1/j\omega C} = 0 \Rightarrow Y(1 + j\omega 3RC) = Zj\omega 3RC$

Sub Z: $Y\frac{(1+j\omega 3RC)}{j\omega 3RC}(1 + 2j\omega RC) - Y(1 + j\omega RC) = Xj\omega RC$

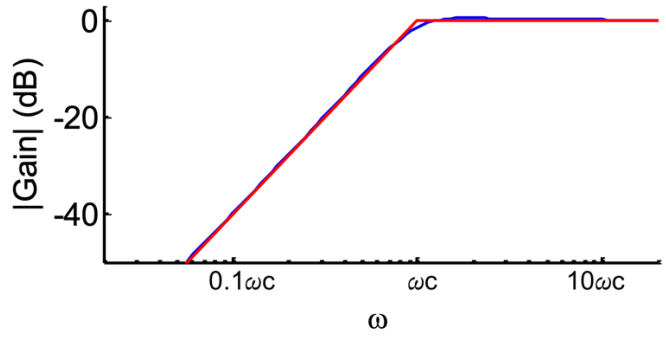
$$\Rightarrow \frac{Y}{X} = \frac{3(j\omega RC)^2}{3(j\omega RC)^2 + 2j\omega RC + 1}$$

Asymptotes: LF = $\left(\frac{j\omega}{\omega c}\right)^2$ *and HF =* 1

Corner freq: $\omega c = \frac{1}{\sqrt{3}RC}$ *rad/s*



Clearly, the circuit is a high pass-filter with the following frequency response.



b) For the circuit shown in *Fig. 2b*, write down the transfer function $H = \frac{Y}{X}(j\omega)$ and sketch a straight-line approximation to the frequency plot of $|H|$ between 10 Hz and 100 kHz. [Only the magnitude plot of $|H|$ is required. You do not need to plot the phase.]

[10]

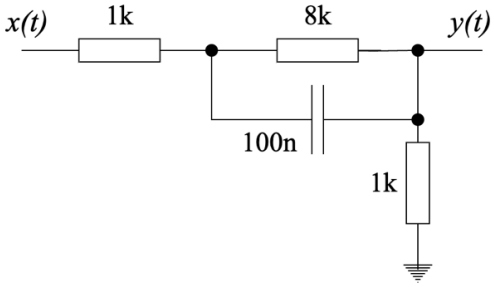


Fig. 2b

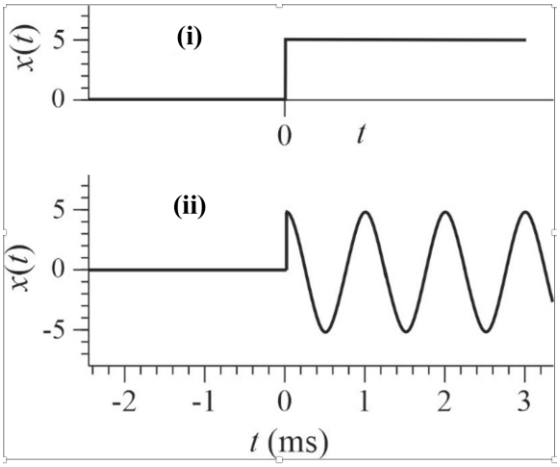


Fig. 2c

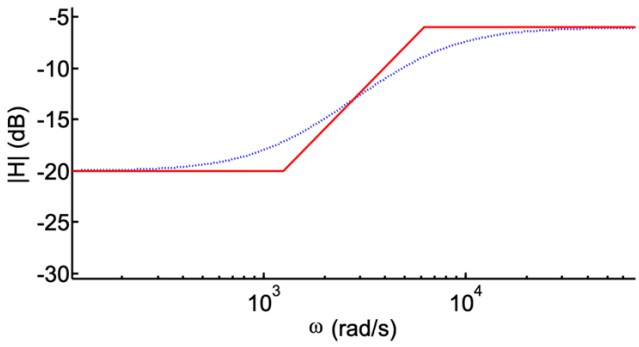
For convenience, we define R = 1k and C = 100n. The total shunt impedance at the output is R. The transfer function is therefore:

$$H = \frac{R}{\frac{8R}{1+j8\omega RC}+2R} = \frac{1+j8\omega RC}{10+j16\omega RC} = 0.1\,\frac{1+j8\omega RC}{1+j1.6\omega RC}$$

RC = 100μs

LF asymptote is 0.1; HF asymptote is 0.5.

Numerator corner frequency is 1/8RC = 1250 rad/s (198.94 Hz) with a gain at the corner of 0.1 = −20dB. Denominator corner frequency is 1/1.6RC = 6.25 krad/s (994.71 Hz) with a gain at the corner of 0.5 = −6dB.
Both the frequency and the gain change by a factor of 5 between the corner frequencies. The frequency plot is given below:



c) For the circuit shown in *Fig. 2c*, obtain an expression for the time variation of the output voltage $y(t)$ when the input signal $x(t)$ is as shown in *Fig, 2c (i)*, with $x(t) = 0$ for $t < 0$ and $x(t) = 5$ for $t \geq 0$.The initial charge on the capacitor is zero. Draw a dimensioned sketch of $y(t)$ covering the time interval $-0.5\text{ ms} \leq t \leq 2\text{ ms}$.

[6]

Method 1 (transfer function): We calculated the transfer function of the circuit in part a). From that we get DC gain as 0.1 and HF gain as 0.5 and time constant as 1.6 RC.

$\tau = 1.6RC = 0.16$ ms

$$\Delta y = G_{HF}\,\Delta x = 0.5 \times 5 = 2.5$$

$$y(t) = y_{ss}(t) + Ae^{\frac{-t}{\tau}}$$

$$A = y(0+) - y_{ss}(0) = 2.5 - 0.5 = 2$$

$$y(t) = y_{ss}(t) + Ae^{\frac{-t}{\tau}} = \begin{cases} 0 & t < 0 \\ 0.5 + 2.0e^{\frac{-t}{\tau}} & t \geq 0 \end{cases}$$

Method 2 (capacitor voltage continuity):

The time constant of the circuit is obtained by setting input voltage as zero and finding the Thévenin's resistance across the capacitor terminals.

$R_{Th} = 2R \,//\, 8R = 1.6R$ giving the time constant, $\tau = 1.6RC = 0.16$ ms

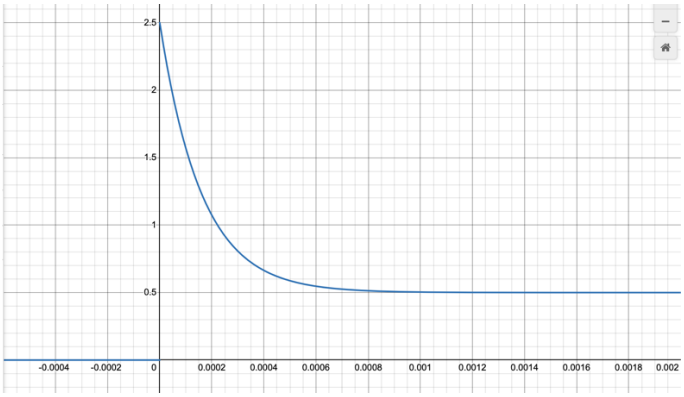
The steady state voltage across the capacitor is given by $v_{ss}(t) = 0.8\,x_{ss}(t) = \begin{cases} 0 & t < 0 \\ 4 & t \geq 0 \end{cases}$

For t < 0, the capacitor voltage is 0. By considering voltage continuity across the capacitor,

$$v(t) = v_{ss}(t) + \big(v(0+) - v_{ss}(0-)\big)e^{\frac{-t}{\tau}} = 4 + (0 - 4)e^{\frac{-t}{\tau}} = 4 - 4e^{\frac{-t}{\tau}}$$

This is the expression for voltage across the capacitor. The output voltage is given by

$$y(t) = \frac{x(t) - v(t)}{2} = = \begin{cases} 0 & t < 0 \\ 0.5 + 2.0e^{\frac{-t}{\tau}} & t \geq 0 \end{cases}$$



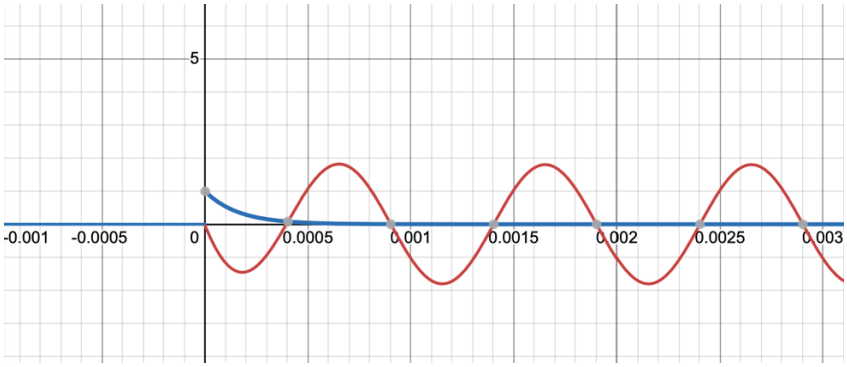
b) For the circuit shown in *Fig. 2b*, using the transfer function method, obtain an expression for the time variation of $y(t)$ when the input signal $x(t)$ is as shown in *Fig. 2c (ii)*, with $x(t) = 0$ for $t < 0$ and $x(t) = 5\cos(2000\pi t)$ for $t \geq 0$.

[6]

The input signal for $t \geq 0$ *has* $\omega = 2000\pi$, *and at this frequency* $\omega RC = \pi/5$, *and the transfer function is* $H = 0.1\frac{1+j8\omega RC}{1+j1.6\omega RC} = 0.3011 + j\,0.200 = 0.361471\angle 33.5934°$. *The steady-state output for* $t \geq 0$ *is therefore* $y_{ss}(t) = 1.807\cos(2000\pi t + 33.60°)$, *and the overall solution for the output voltage is:*

$$y = \begin{cases} 0 & t < 0 \\ 1.807\cos(2000\pi t + 33.60°) + A\,e^{-t/\tau} & t \geq 0 \end{cases}$$

To get the value of A we note that the input step is $\Delta x = +5$, *so the output step must be* $\Delta y = G_{HF}\Delta x = 2.5$. *We therefore require that* $1.807\cos(33.60°) + A = 2.5 \Rightarrow A = 0.995$.



3.

a) Derive a formula for the small signal voltage gain of the common emitter amplifier arrangement of Fig. 3a in terms of V_{CC} and V_T, when the quiescent output voltage is biased at $V_{cc}/2$. The gain is independent of the value of R_C. Why, and what are the implications of this? You may assume the transistor's output resistance is infinite.

[5]

When the output is biased at $V_{CC}/2$, the collector current is:

$V_{CC}/2R_C$

Hence the transconductance, $g_m=I_C/V_T= V_{CC}/2R_C= V_{CC}/(2V_TR_C)$
Hence the gain is $A_V=-V_{CC}R_C/(2V_TR_C)= -V_{CC}/(2V_T)$

Which is independent of R_C. This is because the gain is the product of R_C and g_m, and increasing R_C decreases the collector current which decreases g_m, and vice-versa.

The implication is that (assuming we adhere to the good practice of biasing the output to half of the supply voltage), the gain cannot be adjusted by changing RC, and is only controllable by setting the power supply voltage.

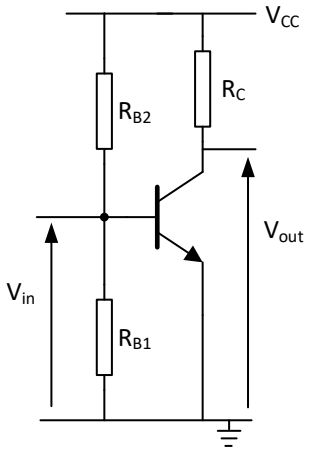


Fig. 3a

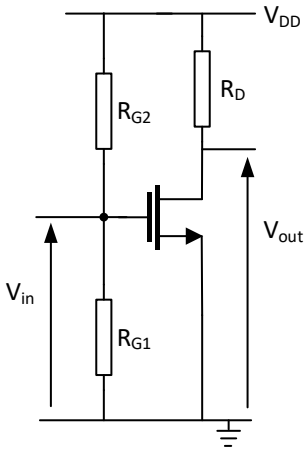


Fig. 3b

b) Derive a formula for the small signal voltage gain of the common drain amplifier arrangement of Fig. 3b), assuming the quiescent output voltage is biased at $V_{DD}/2$. Assume the conductivity parameter of the MOSFET is k_1. Does the magnitude of the small signal voltage gain increase, or decrease with R_D? Assume the transistor output resistance is infinite.

[5]

When the input is biased at $V_{DD}/2$, the drain current is $V_{DD}/2R_D$, which makes the transconductance, $g_m = 2\sqrt{k_1 I_D} = 2\sqrt{k_1 V_{DD}/2R_D}$

And hence the gain, $A_V = -\sqrt{2k_1 R_D V_{DD}}$

In this case it is possible to increase the gain by increasing R_D.

c) Fig 3c shows a MOSFET based differential amplifier.

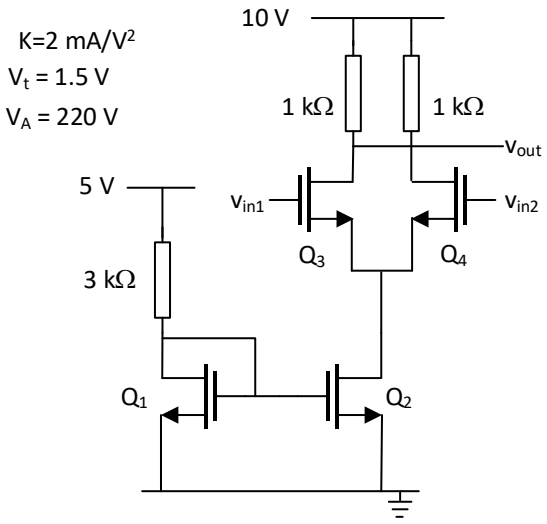
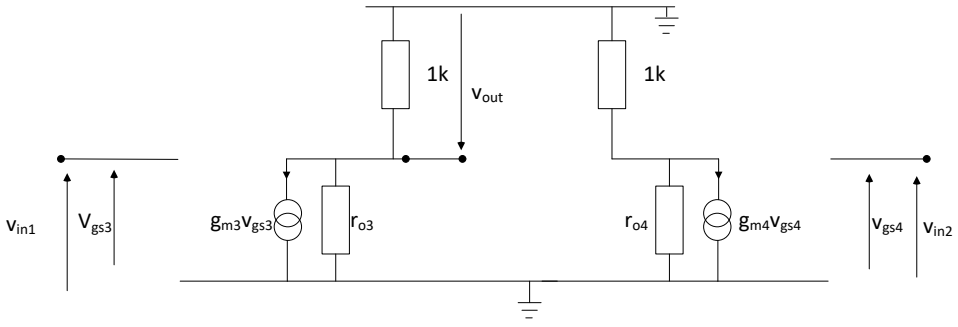


Fig. 3c

i) Draw a small signal model for the circuit in differential mode excitation and calculate the differential voltage gain to the single ended output, V_{out}, when the input voltage is taken as (V_{in1}-V_{in2}). Ensure you have the correct sign.

[8]

The small signal model is:



The Current in Q1 (and hence Q2) is:

$I_D=k(V_{GS}-V_t)^2=2e-3*(V_{GS}-1.5)^2$
and also $I_D=(5-V_{GS})/3$

Solving the quadratic in V_{GS} gives $V_{GS}=2.185$ V or 0.64 V. As Q1 must be turned on $V_{GS}=2.185$ V (as 0.64 V is below the threshold.)
This gives $I_D=0.94$ mA (2 d.p.)

This gives a bias current in Q_3 and Q_4 of 0.47 mA.

The transconductance of Q_3 and Q_4 is thus $2\sqrt{kI_D}$=1.94 mS.

The small signal output resistance r_o of Q_3 and Q_4 is V_A/I_D=220/0.47=468 k.

Thus the differential gain is $-g_m(r_o//R_D)/2$=-0.97.

ii) Calculate the common mode voltage gain of the amplifier and the common mode rejection ratio. You may assume that under common-mode excitation, the change in V_{GS} for transistors Q_3 and Q_4 is negligible.

[4]

The common mode gain is $R_D/2r_{oQ2}$

The output resistance of Q_2 is V_A/I_D=220/0.94=234 k.

*Thus the common mode gain is: 1/(2*234)=0.0021*

And the CMRR is therefore 0.97/0.0021=462

iii) Comment on the performance of this differential gain stage. Is it useful?

[2]

The differential gain of the amplifier seems very low (at just less than 1), but the common mod gain is very low. Therefore the stage can still be used to reject common mode signals.

d) It is proposed to increase the gain of the differential amplifier by connecting the output of *Fig. 3c* to the input of *Fig. 3a* using capacitive coupling.

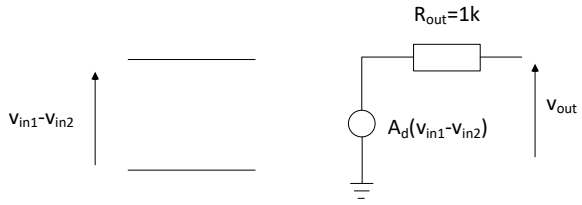
i) Calculate the input and output impedance of Fig 3c and draw a macro model for the amplifier in differential mode.

[3]

The input impedance is effectively infinite (at low to moderate frequency) as the input is straight into a MOS gate. The output impedance is calculated from the small signal model (above) as simply:

*$R_{out}=r_{o3}//R_D$=1*468/(1+468)=0.998kΩ=1kΩ*

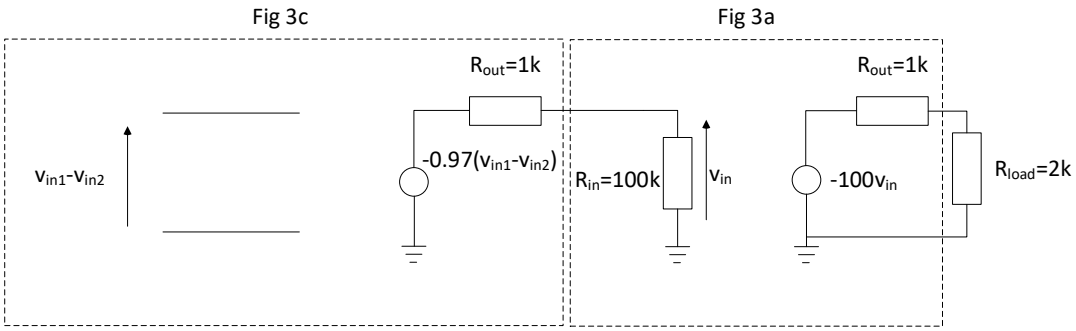
The macro model for differential excitation is thus:



ii) Assume that *Fig. 3a* has been designed with a voltage gain of -100, an input impedance of 100 kΩ and an output impedance of 1kΩ. Draw a macro model of the system where the output of Fig. 3c is capacitively coupled to the input of Fig. 3a, which in turn drives a 2kΩ load. Calculate the mid-band gain.

[3]

The macro model of the system is shown below (in the mid band the coupling capacitance is a short circuit):



Hence the mid band gain is (100/101)(-0.97)*(-100)*(-2/3)=64 (2d.p.)*

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